



Vanderbilt PTY Program

BRIDGING THE GAP: HOW SOCIAL FACTORS CAN GUIDE TYPE 2 DIABETES PREVENTION



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BACKGROUND

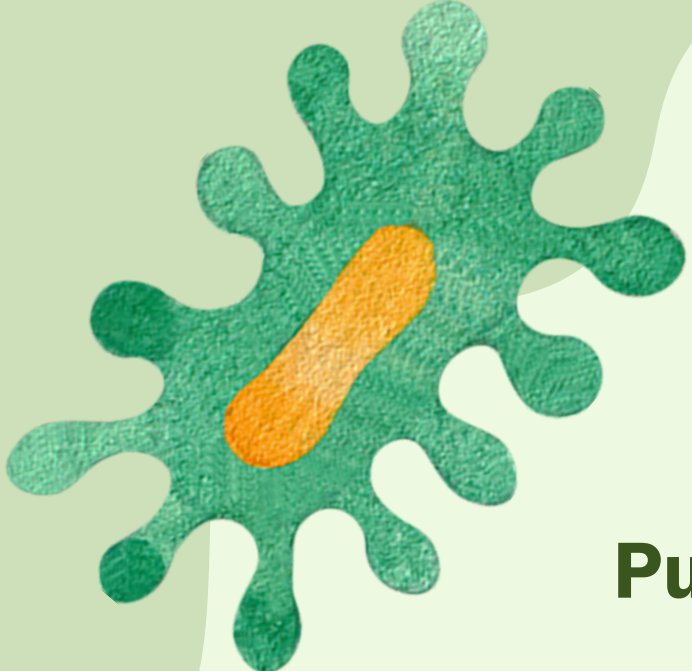
Diabetes in America

- Almost 40 million Americans have type 2 diabetes
- Out-of-pocket cost for diabetes management around \$4,000 annually
- 7th leading cause of death in America

“Social Determinants of Health (SDOH) have been shown to have a greater influence on health than either genetic factors or access to healthcare services”
- CDC

Why do SDOH even matter for Type 2 Diabetes

- Social factors can affect diet
- Living environment can affect exercise frequency
- Poor education leads to inadequate health literacy
- Tough living conditions can increase chronic stress



RESEARCH PROBLEM

Purpose of Study

- Most current diabetes risk models prioritize clinical factors
- Not a lot of information on how SDoH can lower diabetes odds
- This project uses multivariable logarithmic regressions to measure the impact of social factors while controlling for demographic factors

Research Question

Among US adults aged 35-70 in 2017-2018, how are poverty income ratio and education level independently associated with the prevalence of diabetes?



METHODOLOGY (TOOLS)

NHANES

- Primary dataset used
- Sample size of 3,336 patients
- Variables used: diabetes status, educational attainment, poverty-income ratio, gender, ethnicity

Software

- Data processing conducted using Python, Pandas, SciPy, StatsModels and NumPy

```
#INDFMPIR - poverty to income ratio
#DMDEDUC2 - education level
#DIQ010 - diabetes status
#RIDRETH3 - race
#RIAGENDR - biological sex
#RIDAGEYR - age

#data-binning for age
#histogram for age and sex
#chi-squared analysis on cross-tabs
import pandas as pd
from scipy.stats import chi2_contingency
import statsmodels.formula.api as smf
import numpy as np

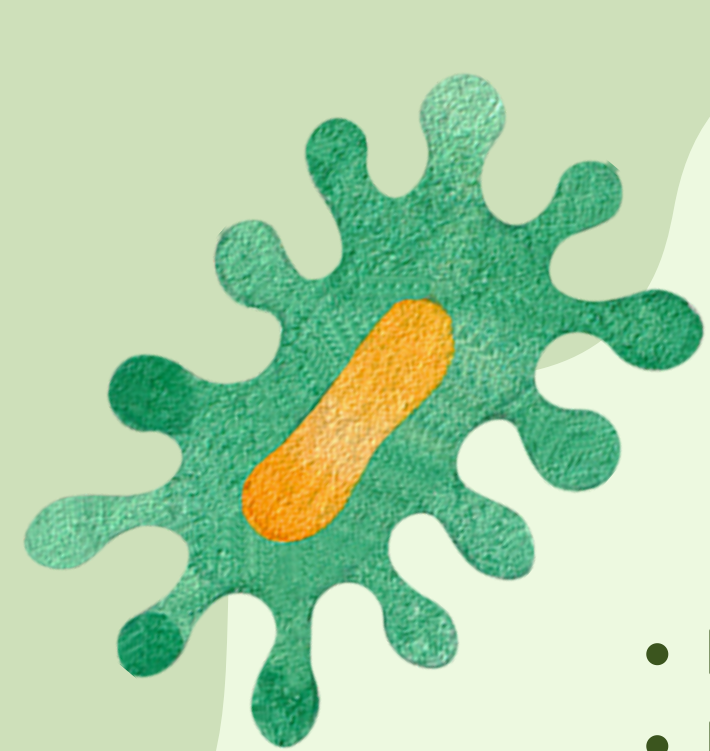
demo_df = pd.read_sas('/content/DEMO_J.xpt', format='xport', encoding='utf-8')
diq_df = pd.read_sas('/content/DIQ_J.xpt', format='xport', encoding='utf-8')

merged_df = pd.merge(demo_df, diq_df, on='SEQN', how='inner')

#all variables
raw_vars = ['INDFMPIR', 'DMDEDUC2', 'DIQ010', 'RIDRETH3', 'RIAGENDR', 'RIDAGEYR']

#Filter for age
df_study = merged_df[(merged_df['RIDAGEYR'] >= 35) & (merged_df['RIDAGEYR'] <= 70)][raw_vars].copy()

#Clean data
df_study['DMDEDUC2'] = df_study['DMDEDUC2'].replace([7.0, 9.0], None)
df_study['DIQ010'] = df_study['DIQ010'].replace([7.0, 9.0], None).map({1: 1, 2: 0})
```



METHODOLOGY CONTD



- Data from 2017-2018 NHANES cycle used
- Dataset merged and cleaned by removing unreported/missing values
- Grouped income into brackets
- Descriptive statistics and cross tabulations run first
- Chi squared analysis run on the two SDoH factors
- Logarithmic regressions run to account for demographic factors

```
# Logistic regression for PIR
model_pir = smf.logit(
    "DIQ010 ~ INDFMPIR + C(RIDRETH3) + C(RIAGENDR)",
    data=df_study
).fit()

results_pir = pd.DataFrame({
    "Odds Ratio": np.exp(model_pir.params),
    "Lower 95% CI": np.exp(model_pir.conf_int()[0]),
    "Upper 95% CI": np.exp(model_pir.conf_int()[1]),
    "p-value": model_pir.pvalues
})

print(results_pir)

incomechi = chi2_contingency(cross_income_diab)
print("\n\nIncome vs diabetes chi squared analysis")
print(incomechi)
```

Logistic Regression

```
#Cross-tab comparing income brackets vs diabetes
print("\nCross-Tab: Income Bracket vs. Diabetes Diagnosis")
cross_income_diab = pd.crosstab(
    df_study['income_bracket'],
    df_study['DIQ010'],
)
display(cross_income_diab.round(2))

#chi squared analysis for income-poverty ratio
incomechi = chi2_contingency(cross_income_diab)
print("\n\nIncome vs diabetes chi squared analysis")
print(incomechi)
```

Chi Squared Analysis



RESULTS

- Overall diabetes presence: 17.6%
- Higher education correlates with lower diabetes rate ($X^2=27.88$, $p<0.001$)
- Diabetes prevalence drops from 19.5% in lowest income bracket to 14.9% in highest income bracket
- Females consistently had lower rates of diabetes than males

Optimization terminated successfully.
Current function value: 0.459396
Iterations 6

	Odds Ratio	Lower 95% CI	Upper 95% CI	p-value
Intercept	0.414345	0.307924	0.557547	5.981606e-09
DMDEDUC2 [T.2.0]	0.682586	0.470939	0.989350	4.374700e-02
DMDEDUC2 [T.3.0]	0.570932	0.406884	0.801121	1.182901e-03
DMDEDUC2 [T.4.0]	0.611336	0.441209	0.847064	3.101819e-03
DMDEDUC2 [T.5.0]	0.419959	0.294513	0.598840	1.649370e-06
C(RIDRETH3) [T.2.0]	0.801057	0.554689	1.156852	2.368281e-01
C(RIDRETH3) [T.3.0]	0.864615	0.634716	1.177784	3.563213e-01
C(RIDRETH3) [T.4.0]	1.025440	0.749915	1.402196	8.749661e-01
C(RIDRETH3) [T.6.0]	1.076914	0.758701	1.528592	6.783918e-01
C(RIDRETH3) [T.7.0]	1.234157	0.780550	1.951373	3.680941e-01
C(RIAGENDR) [T.2.0]	0.806826	0.671722	0.969103	2.169804e-02

Education logarithmic regression

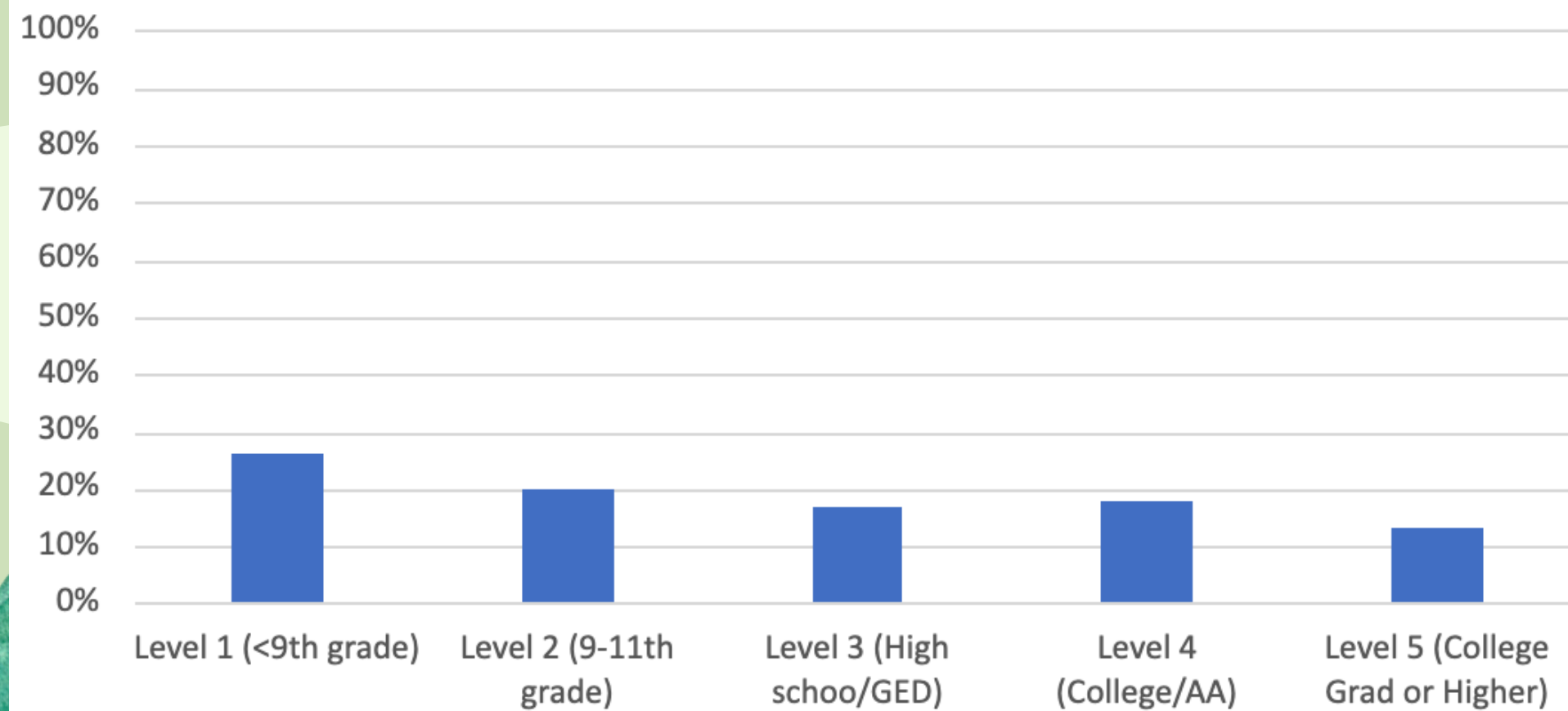
Optimization terminated successfully.
Current function value: 0.454676
Iterations 6

	Odds Ratio	Lower 95% CI	Upper 95% CI	p-value
Intercept	0.373871	0.277157	0.504335	1.178383e-10
C(RIDRETH3) [T.2.0]	0.743618	0.498370	1.109553	1.468305e-01
C(RIDRETH3) [T.3.0]	0.681135	0.499756	0.928344	1.507378e-02
C(RIDRETH3) [T.4.0]	0.810479	0.588793	1.115633	1.974585e-01
C(RIDRETH3) [T.6.0]	0.775302	0.538241	1.116773	1.716812e-01
C(RIDRETH3) [T.7.0]	0.876551	0.540811	1.420720	5.928200e-01
C(RIAGENDR) [T.2.0]	0.746301	0.611655	0.910586	3.943034e-03
INDFMPPIR	0.924033	0.867674	0.984054	1.387095e-02

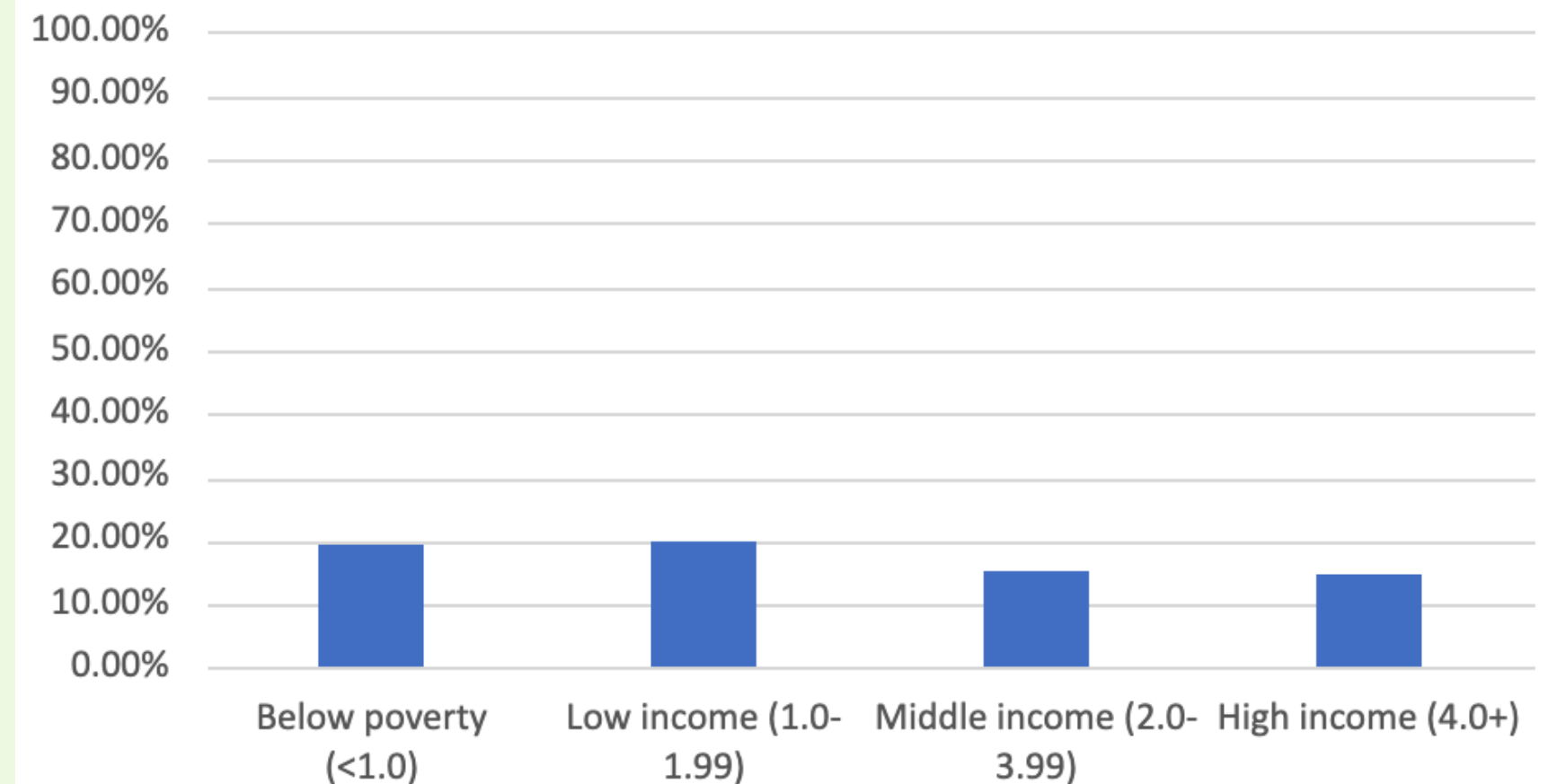
Poverty-Income Ratio logarithmic regression

RESULTS CONTD

Diabetes Prevalence by Education Level



Diabetes prevalence by income bracket





DISCUSSION

We need to protect the most vulnerable members of our community



Including SDoH in CDSS will allow clinicians to more easily identify high-risk patients



Accounting for SDoH reduces bias in AI machine learning models

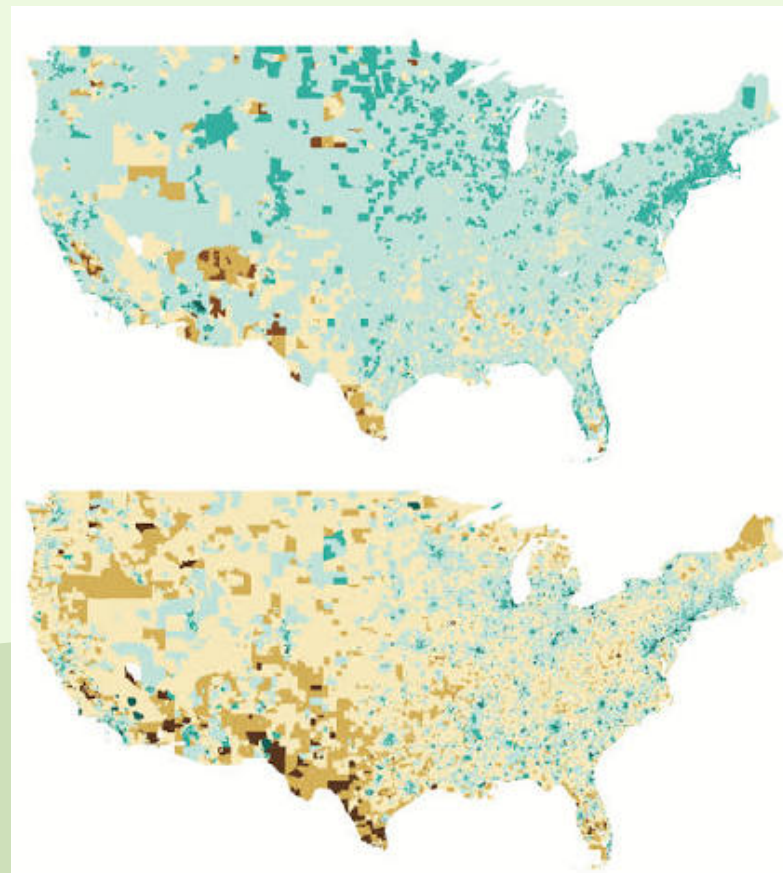


Enables clinicians to trigger community health referrals automatically for patients with risk flags



PUBLIC HEALTH POLICY

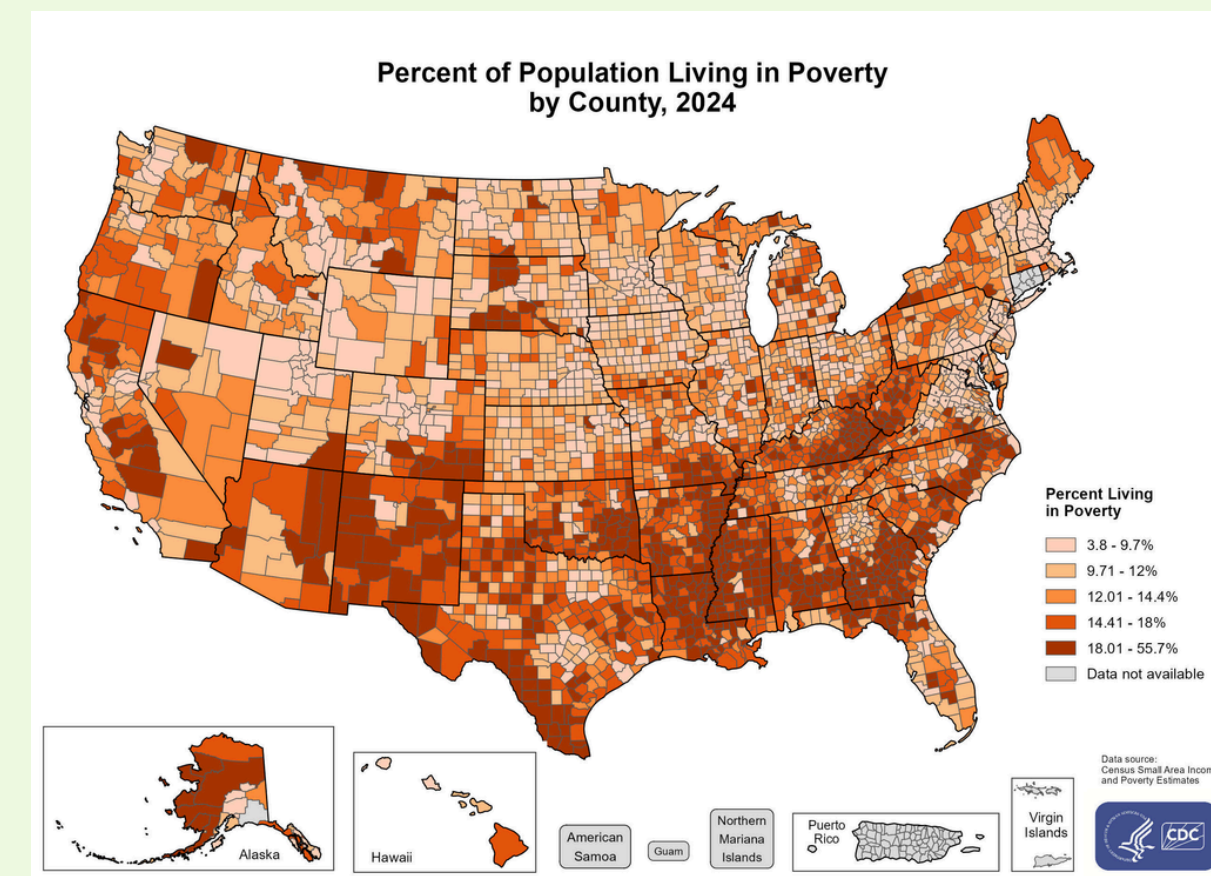
- SDoH maps can allow public health agencies to find low-education and low-income areas and set up mobile clinics, nutritional support and diabetes screenings
- Subsidized DSME programs for lower-income groups
- Diabetes prevention requires work **outside** the hospital



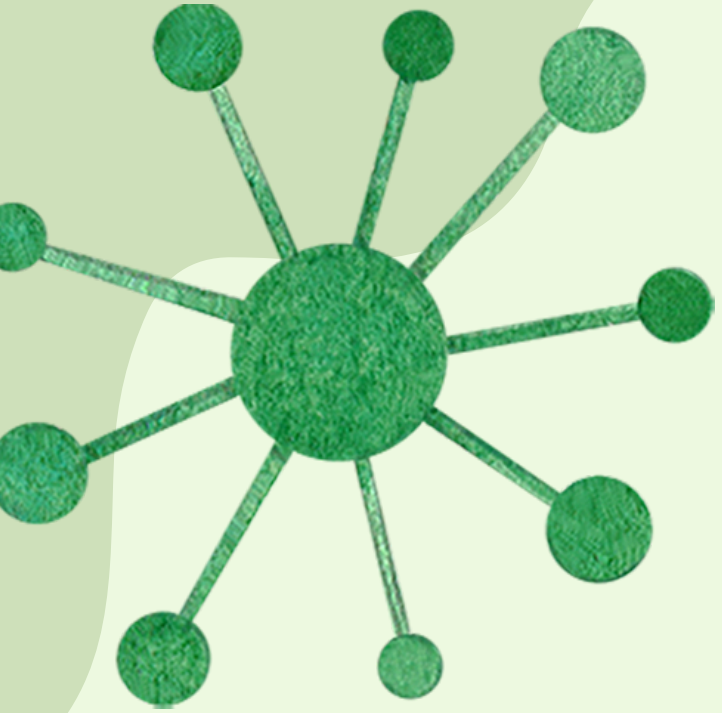
The SDOH Atlas



Diabetes Self-Management Education



Poverty map of United States



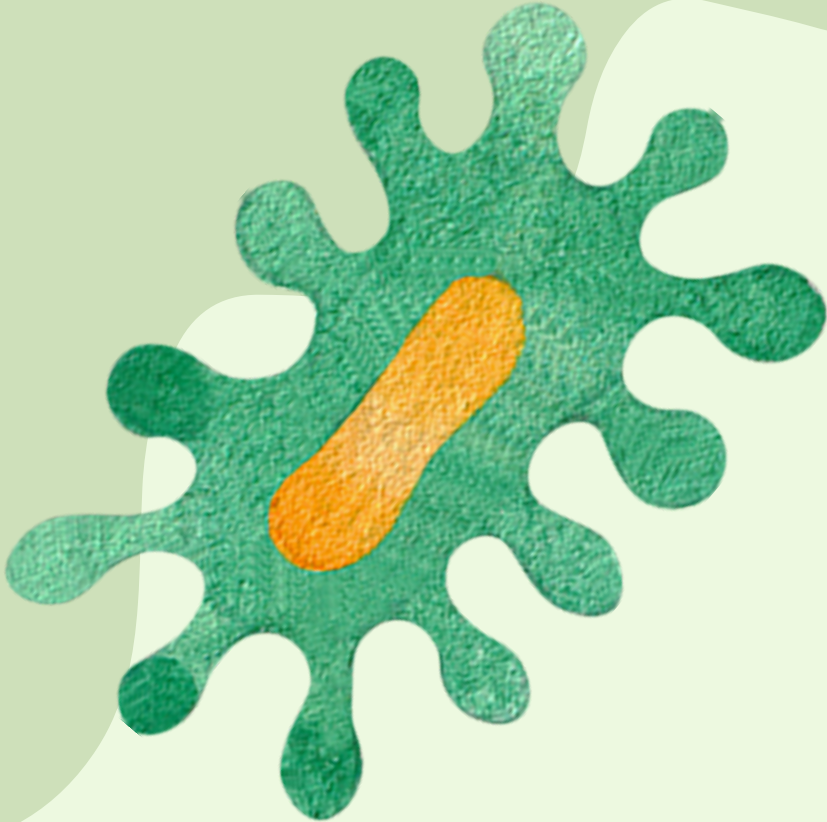
LIMITATIONS AND FUTURE WORK

Limitations

- Can only definitively establish correlation
- Many SDoH variables not included
- Diabetes Variable (DIQ010) relies on self-reported diabetes diagnoses

Future Work

- Add machine learning to electronic health records to track how changes in SDoH affects diabetes prevalence over time
- Combine SDoH variables with lab variables like HbA1c to build more effective risk-models for diabetes



CONCLUSION & KEY TAKEAWAYS



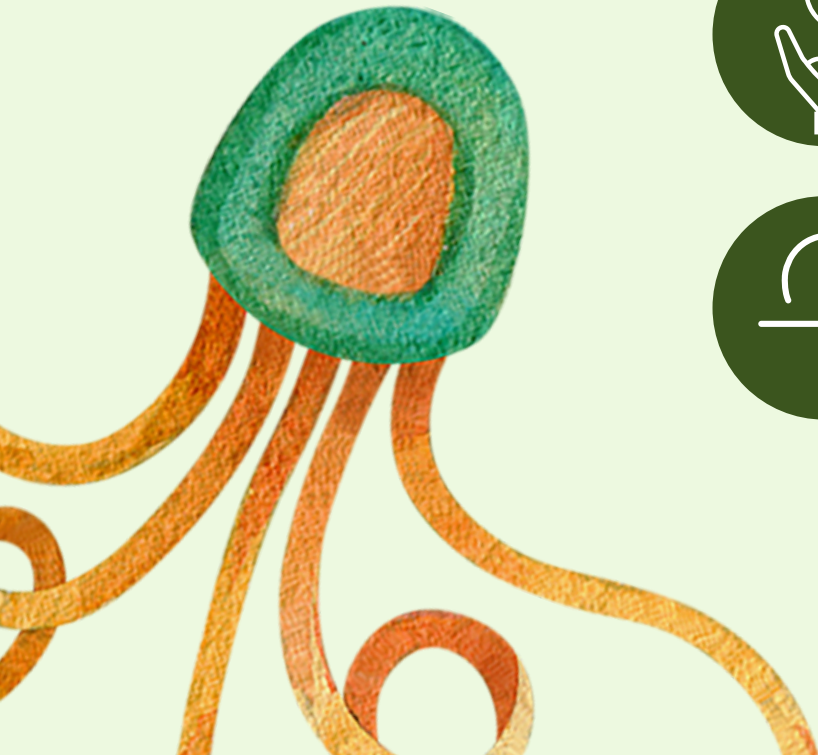
SDoH factors can be just as important as clinical factors in determining diabetes risk



Higher poverty-income ratios and education levels significantly reduce diabetes risk



Clinical risk models must incorporate SDoH factors and we should use public health datasets like NHANES to design interventions for high risk groups



ACKNOWLEDGEMENTS

- My parents
- Dr. Alex Becker
- NHANES



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**THANK
YOU**

